

	Work Instruction	Doc. Revision	03
	AXI V810 Y Axis Ball Screw Alignment Procedure	Effective Date	20 Oct 2016

This procedure is applicable to below models:

V810 STD	☒	S1	☒	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	30 Jan 2013	First Release	ONG KEAN SHENG
01	04 Jan 2015	Second Release	MUHAMMED SHAMEEM
02	01 Feb 2015	Picture revise	ONG KEAN SHENG
03	20 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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1. Loosen the bearing the Bearing support with M6 Allen Key at “A” & “B” (Note: DO NOT TIGHTEN) as shown in Figure 1.
2. Loosen the Bearing support, Fixed round at M5 Allen Key at “G” & “H”. (Note: DO NOT TIGHTHEN) as shown in Figure 1.
3. Loosen the screw at round bearing with M6 Allen Key at “C” to “F”. (Note: DO NOT TIGHTEN) as shown in Figure 1.

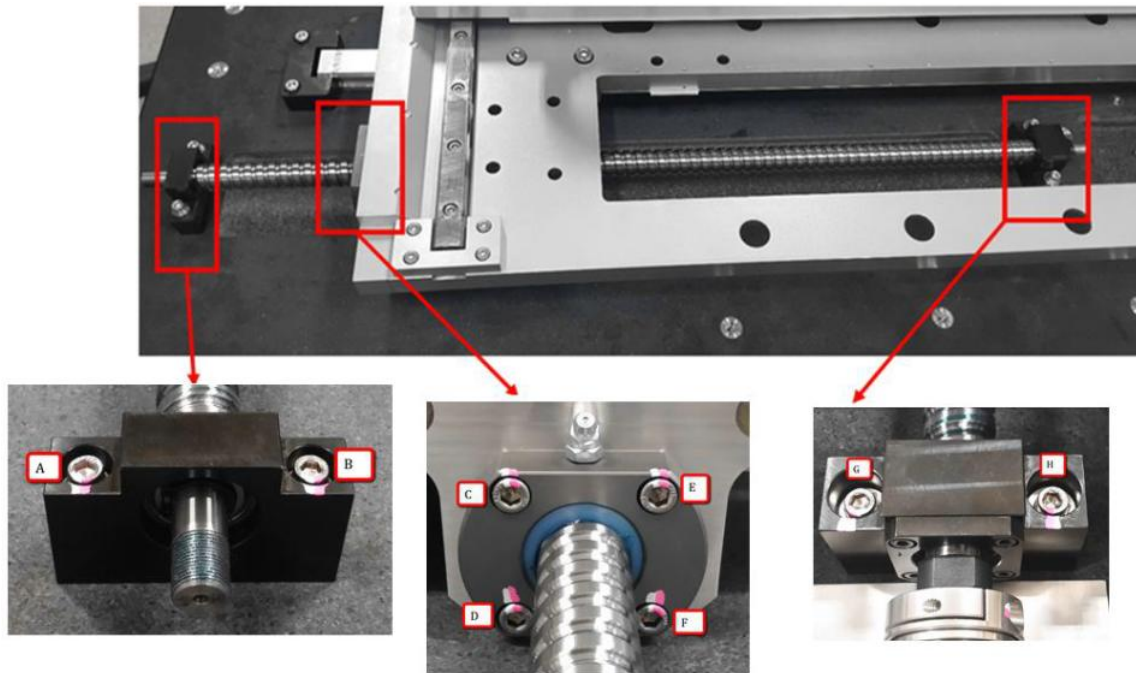


Figure 1 Y-Axis Ball Screw

4. Loosen the 2 coupling screw of which is attached to the Y-Axis Ball screw and Y-axis motor as shown in the Figure 2.

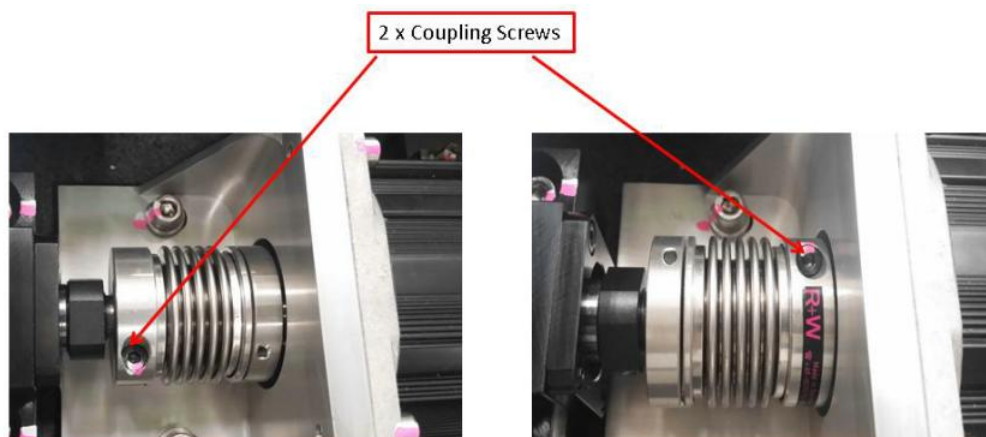


Figure 2 Coupling Screws

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- Loosen the 2 screw on the motor mount as shown in Figure 3.

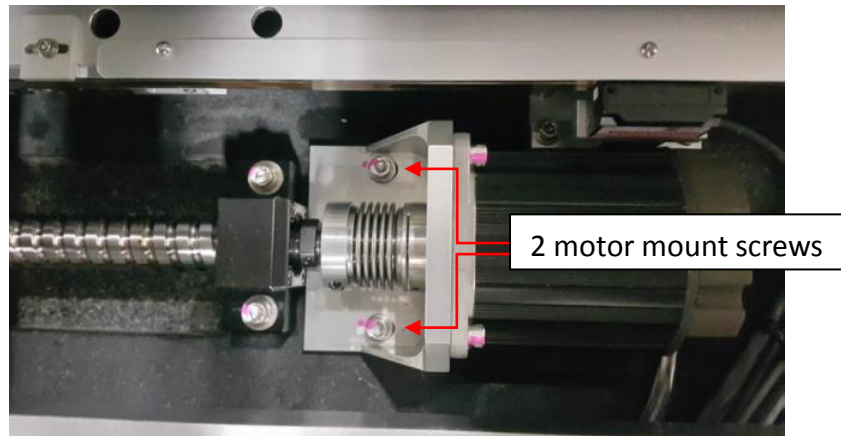


Figure 3 Motor Mount

- Push the stage from minimum to maximum hard stop (Front to Rear)
- Pull the stage from maximum to minimum hard stop (Rear to Front)
- Repeat Step 6 to 7 for 3 times.

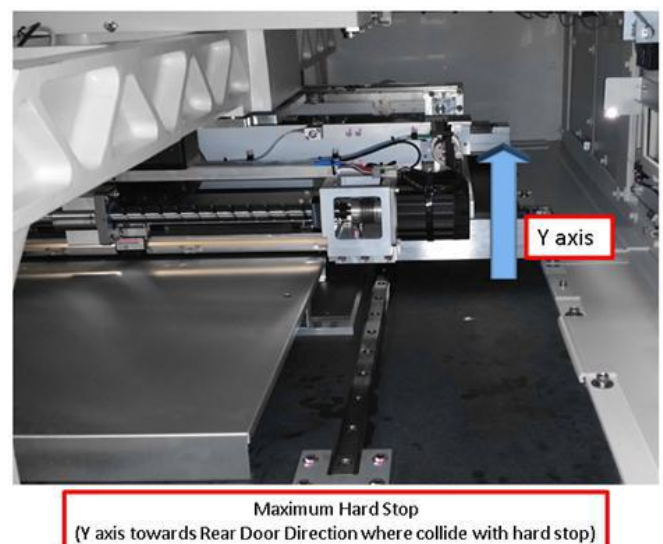
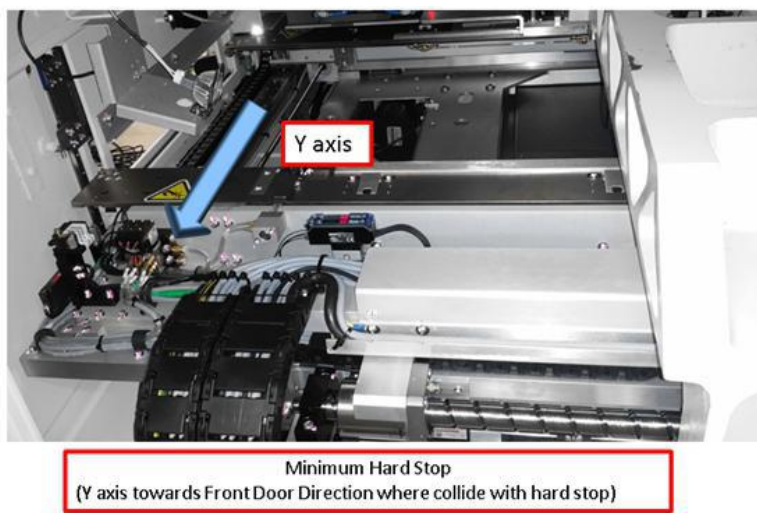


Figure 4 Minimum & Maximum Hard Stop (V810 Std)



Figure 5 Minimum & Maximum Hard Stop (V810 XXL)

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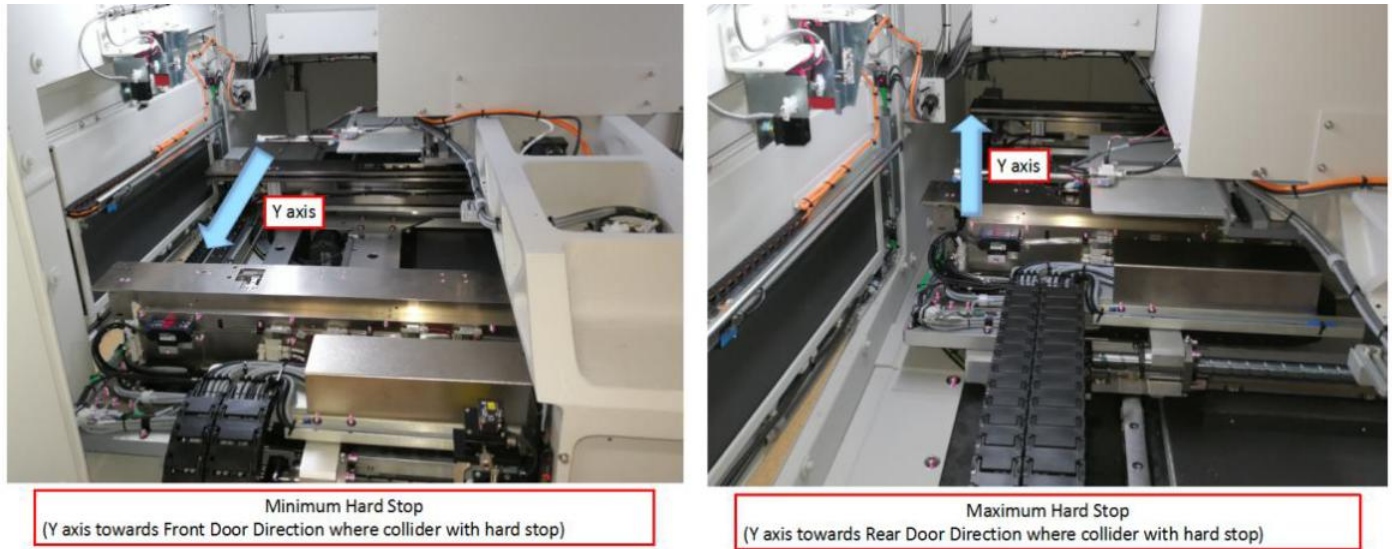


Figure 6 Minimum & Maximum Hard Stop (V810 S2EX)

- Push the stage to the middle as shown in the Figure 7. Tighten the screw "C" to "F".

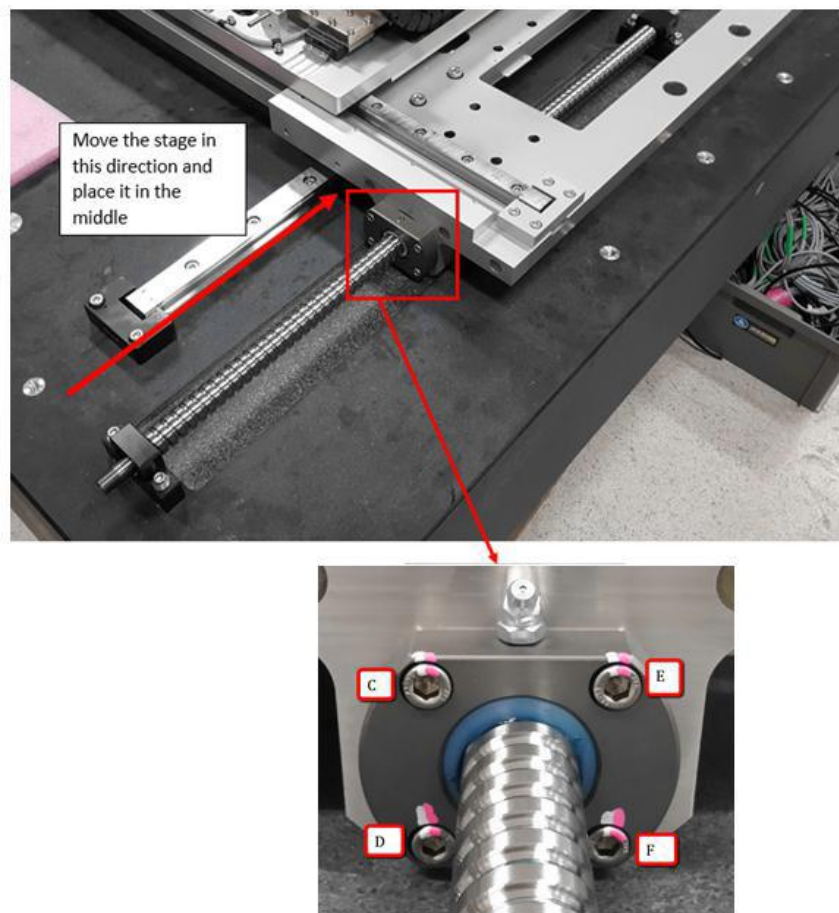


Figure 7 Y-Axis Round Bearing

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10. Push the stage from minimum to maximum hard stop (Front to Rear)
11. Pull the stage from maximum to minimum hard stop (Rear to Front)
12. Repeat Step 10 to 11 for 3 times.

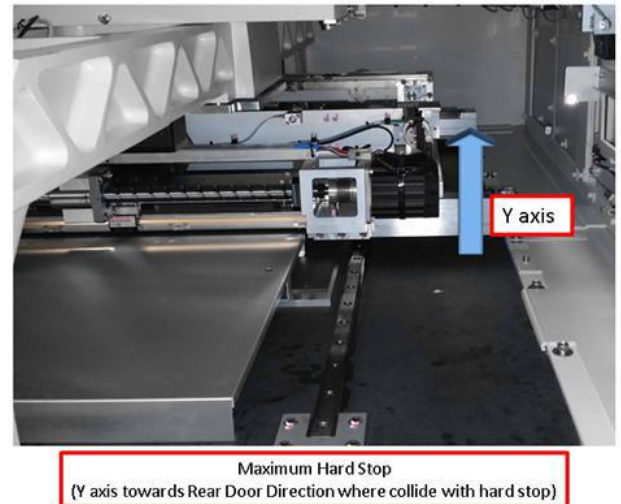
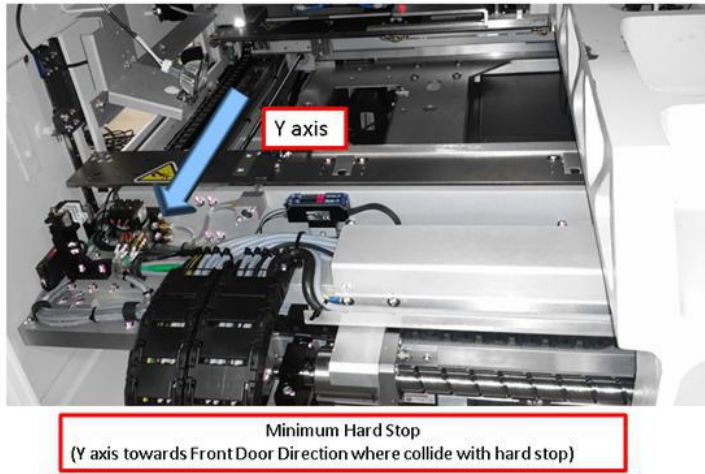


Figure 8 Minimum & Maximum Hard Stop (V810 Std)

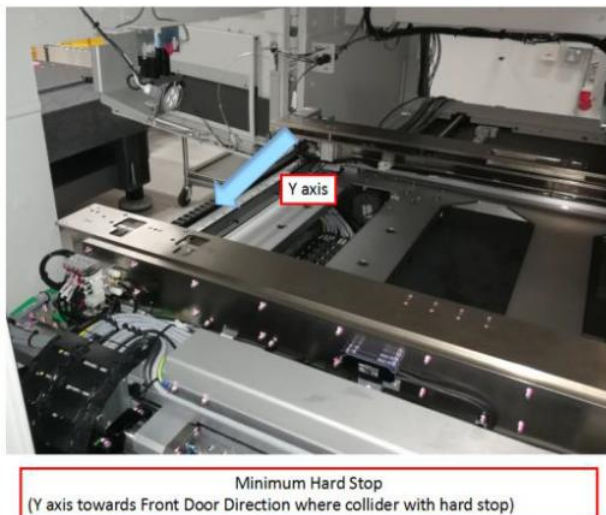


Figure 9 Minimum & Maximum Hard Stop (V810 XXL)



Figure 10 Minimum & Maximum Hard Stop (V810 S2EX)

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13. Move the stage to the minimum hard stop and tighten and tighten the screw “G” & “H” as shown in the Figure 11.

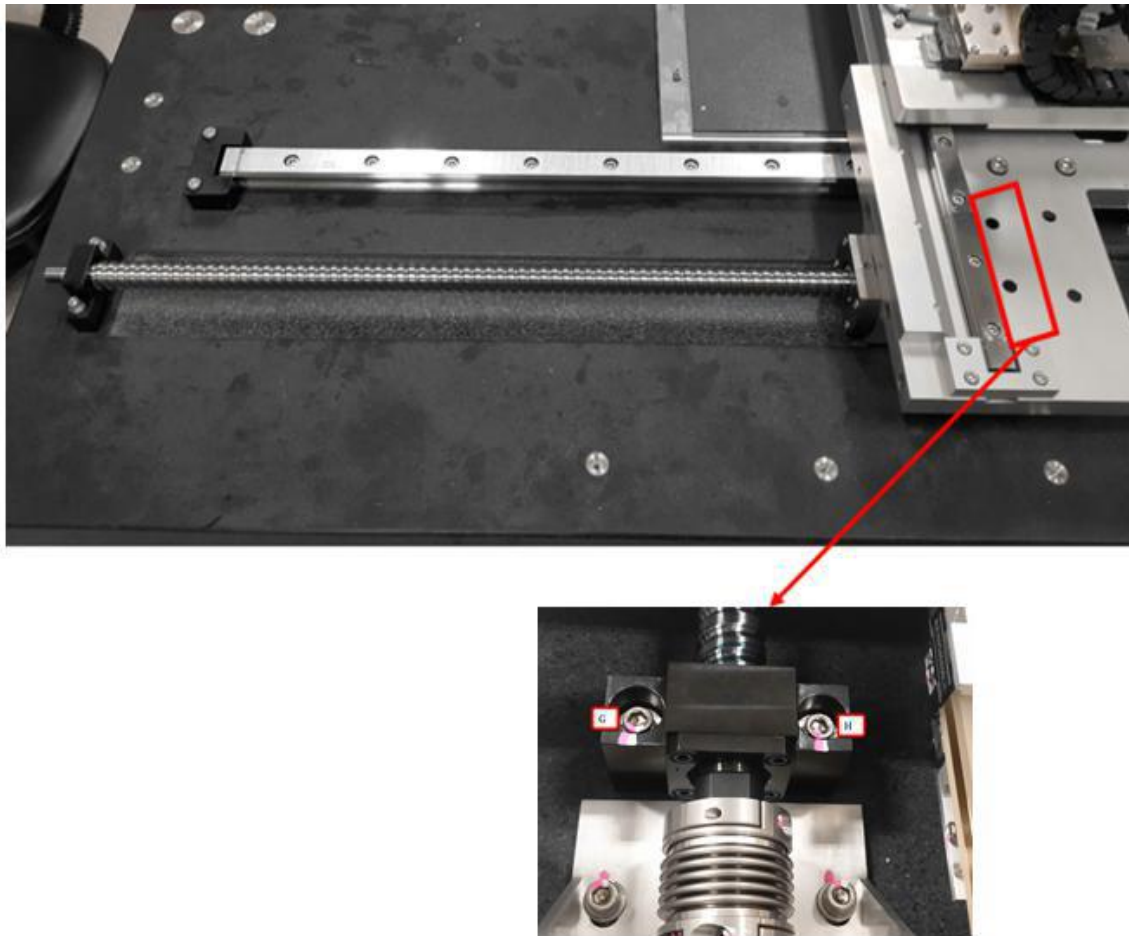


Figure 11 End Bearing Support

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14. Push the stage from minimum to maximum hard stop (Front to Rear)
15. Pull the stage from maximum to minimum hard stop (Rear to Front)
16. Repeat Step 14 to 15 for 3 times.

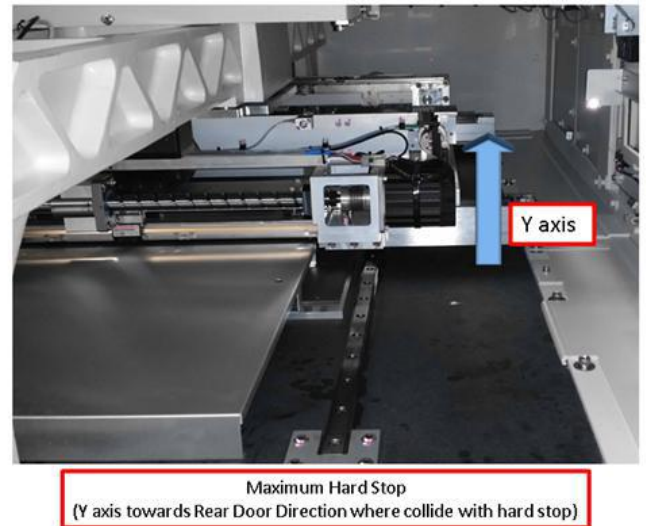
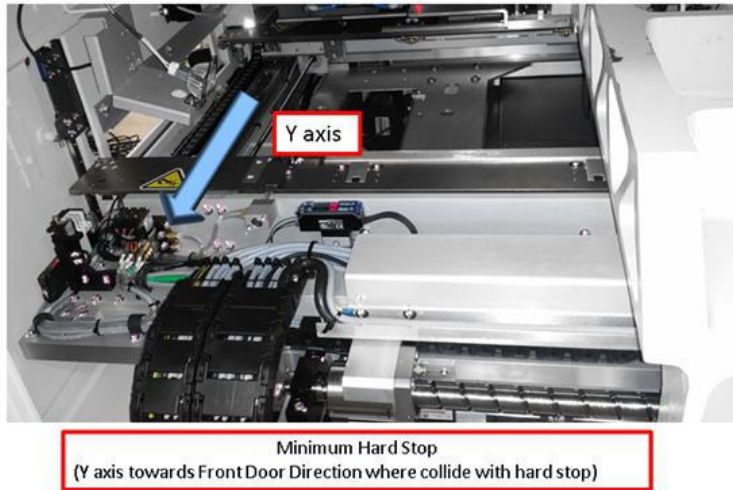


Figure 12 Minimum & Maximum Hard Stop (V810 Std)

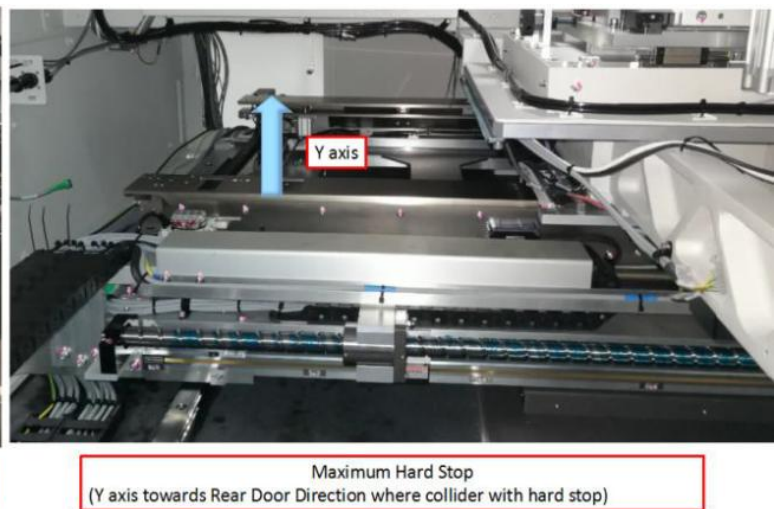
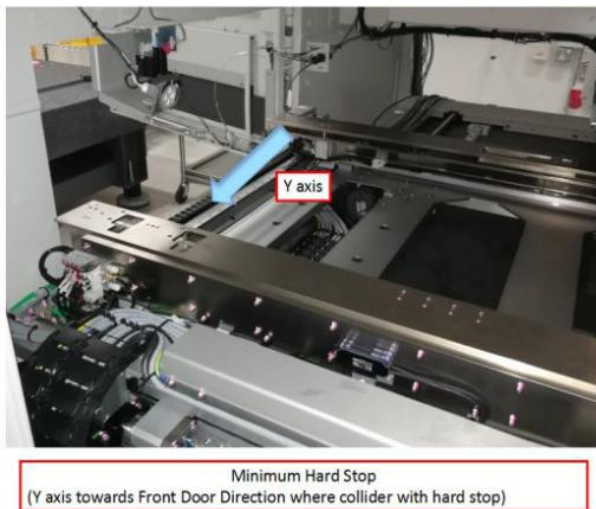


Figure 13 Minimum & Maximum Hard Stop (V810 XXL)



Figure 14 Minimum & Maximum Hard Stop (V810 S2EX)

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17. Move the Y-Axis stage towards the maximum hard stop as shown in the Figure 15 and tighten screw “A” & “B”.

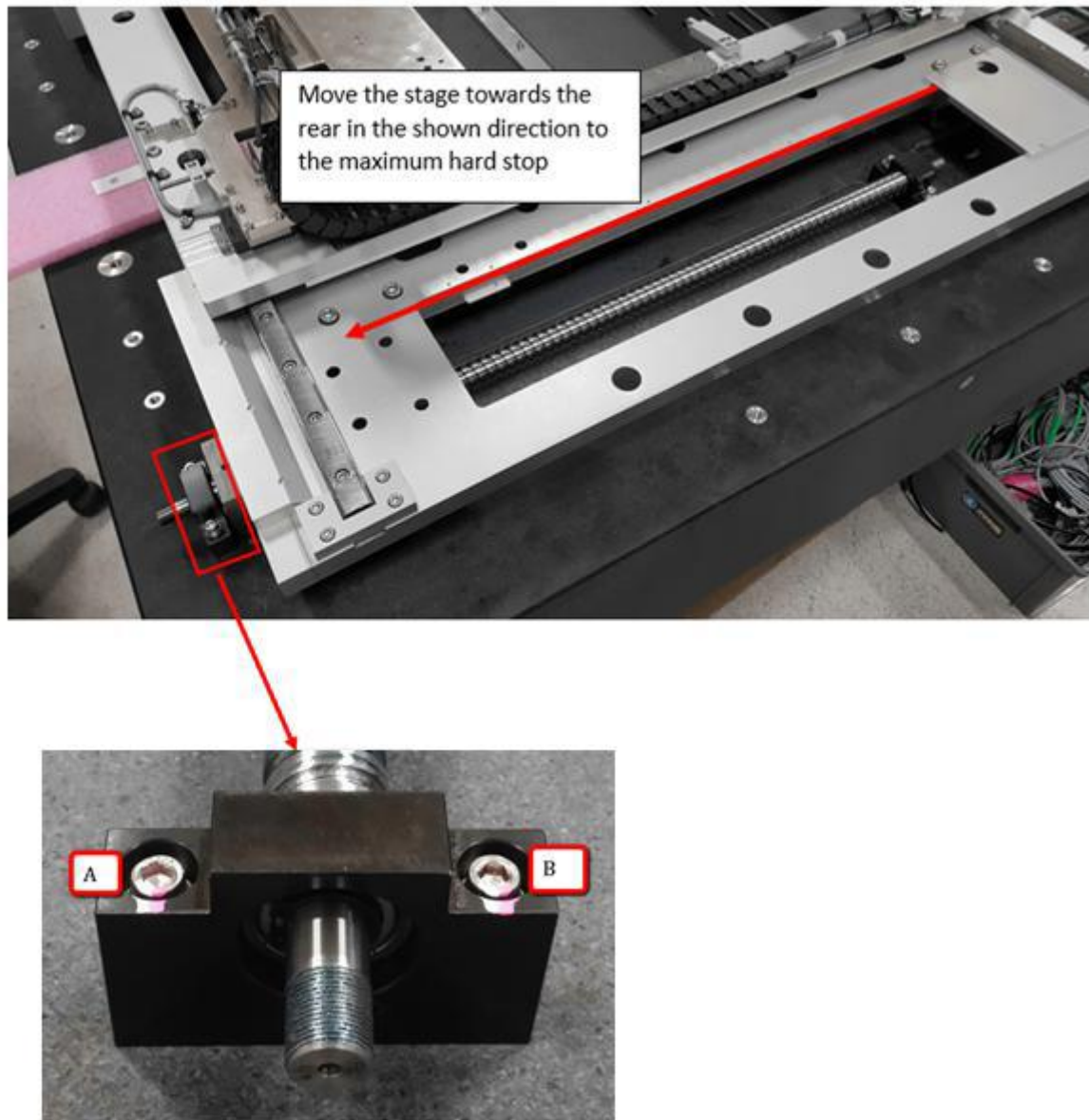


Figure 15 End Support Bearing

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18. Push the stage from minimum to maximum hard stop (Front to Rear)
19. Pull the stage from maximum to minimum hard stop (Rear to Front)
20. Repeat Step 18 to 19 for 3 times.

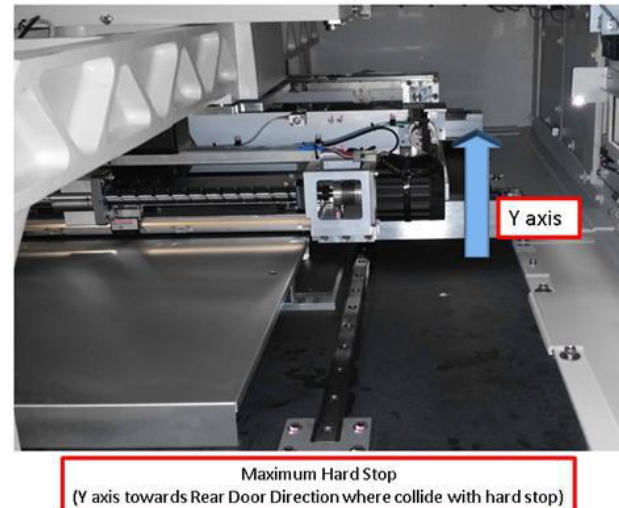
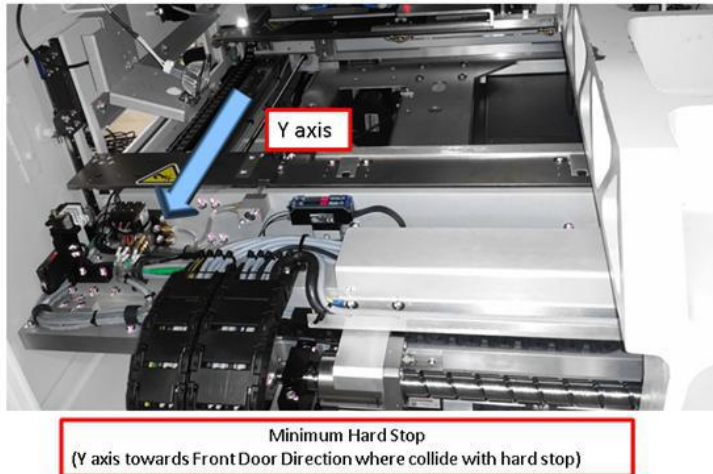


Figure 16 Minimum & Maximum Hard Stop (V810 Std)

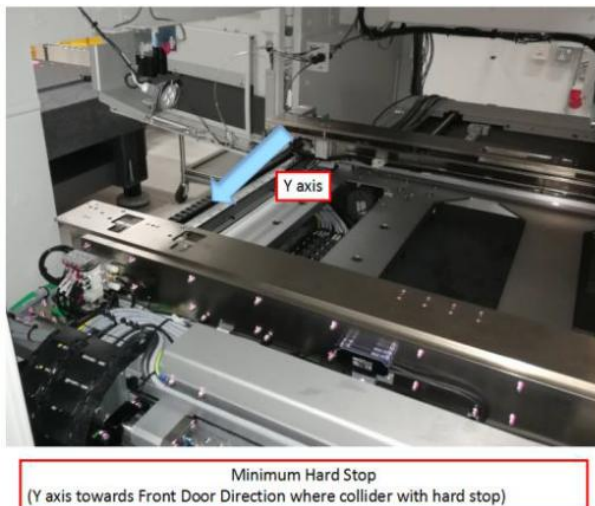


Figure 17 Minimum & Maximum Hard Stop (V810 XXL)

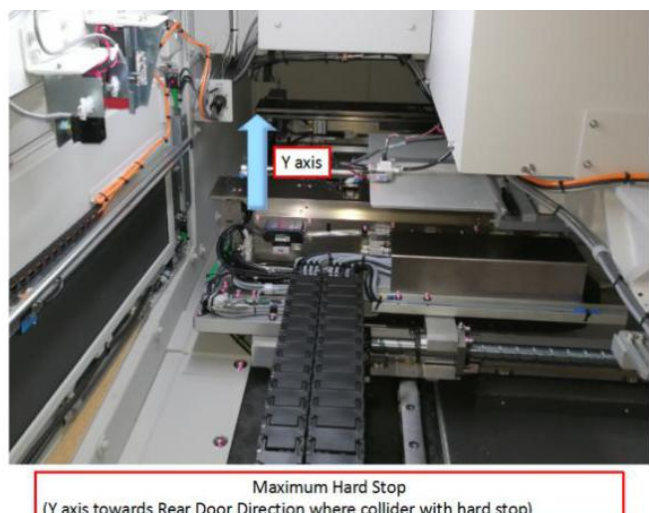


Figure 18 Minimum & Maximum Hard Stop (V810 S2EX)

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Tighten back 2 x Motor Mount screws as below Figure 19.

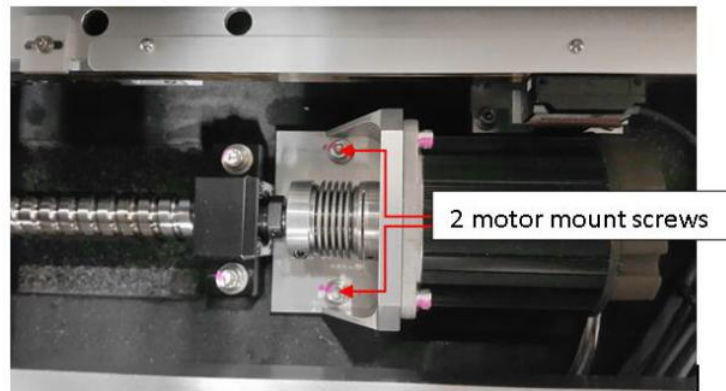


Figure 19 Motor Mount Screws

21. Tighten back 2 x coupling screws as below Figure 20.

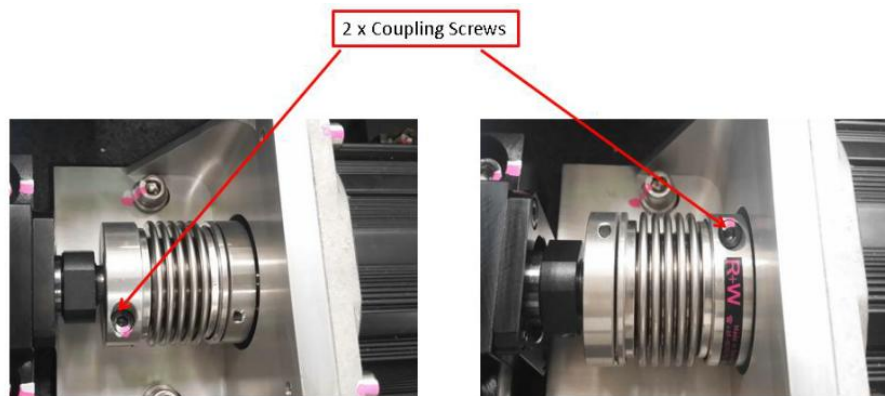


Figure 20 Coupling Screws

22. Move the stage to the rear of the machine & close the front access door.
23. Go to V810 software and log in accordingly.
24. Go to the Services->Subsystem Status & Control -> Panel Positioner and Click Initialize.
25. Run 5 Loops of Long Scan at Panel Positioner.
26. Run 1 loop of CD &A.
27. If Y axis stage is encounter abnormality during step 26 Or 27, repeat Step 1 top Step 22.